

Fairness in Model Compression

Ethan Epp, Jasper Ou, Torin Schlunk, Jack Shoemaker

I. INTRODUCTION

Over the years deep neural networks have grown from a few million parameters to upwards of a trillion or more. As a result, these models require significantly more computational power, memory, and energy to operate, beginning to approach the upper limit of available resources. These resource demands pose a key challenge in the operation of these models, especially in their deployment on edge and mobile devices. Therefore there is an ever increasing interest in model compression techniques due to their reduction in models' resource demands. Compressing models is known to have an associated cost in the overall model performance. However, the effects of model compression on fairness is a relatively under-explored field and in this work we aim towards providing a comprehensive overview of its possible effects across a set of different fields.

II. RELATED WORK

Our work builds on prior attempts to characterize the impact of model compression on fairness. Hooker et al. [1] pointed out that compression techniques that evaluate well by accuracy may disproportionately affect accuracy on minorities; testing a classification neural network on CelebA, they showed that magnitude based compression and 8-bit quantization had the potential to disproportionately affect different subgroups. Stoychev and Gunes [2] tested 8-bit quantization and pruning with post-training on facial expression classification, and found that quantization had a negligible impact but that their pruning method could significantly reduce fairness in some cases. Fairness-aware pruning methods such as FairGRAPE [3] have been proposed to reduce unfairness introduced by pruning.

Model compression is of particular interest in applications that call for especially large models, such as natural language processing (NLP). However, measuring fairness in NLP can be challenging. Xu and Hu [4] found that distillation and pruning could actually improve fairness performance of BERT and GPT2. Further experiments by Ramesh et al. [5] had conflicting results, showing an increase of bias when distilling or pruning BERT, but possible benefits from quantization.

Effects of compression on fairness in automatic speech recognition (ASR) are not widely explored, despite ASR models often being rather large and speedups being highly desirable. One publication by Ferraz [6] found that quantization did not have much of an effect on accuracy between genders, but did have a significant disproportionate effect on some languages in the multilingual Whisper models. Ferraz trained a distilled Whisper model which was better able to retain multilingual performance.

In addition to compressing regular models, we also wanted to investigate how compression affects models with a fairness-aware training process. Zafar et al. [7] formulate fairness as constraints that can be directly incorporated into the optimization objective during model training. Their methodology introduces decision boundary covariance constraints that can enforce various fairness criteria, including demographic parity and equalized odds, without significantly compromising accuracy. We attempt to observe how the achieved training fairness by this method is affected by post-compression.

III. MODEL COMPRESSION OVERVIEW

Model compression is the general term given to methods directed towards the reduction in size, computational complexity, and power consumption of machine learning models. These techniques are primarily applied to deep neural networks because of their large sizes and ease of compressing. Over the years there have been a multitude of methods proposed for model compression. However in our work we focus on three of the most widely used methods which are pruning, distillation (or knowledge distillation), and quantization. These compression methods have seen widespread use in a variety of applications, which is further supported by their availability in many popular machine learning frameworks such as PyTorch and TensorFlow.

Distillation, also referred to as knowledge distillation, is a technique where a smaller, more compact model, is trained to replicate the behavior and predictions of a larger and more complex model. Seminal work by Hinton et al. showed that distillation can work well for transferring knowledge from a large highly regularized model into a smaller, distilled variant [8]. The general process involves taking the knowledge from the larger (teacher) model and transferring it to the smaller (student) model. Usually the student model will be a scaled down version of the teacher model, with both fewer and less wide layers. Knowledge transfer can be achieved through training the student model on a combination of the training data and the teacher's output logits. Through this approach the student model is able to achieve similar levels of performance to the teacher model whilst also being smaller and thus computationally more efficient. DSTIL-BERT for example is 40% smaller than BERT however it achieves roughly 97% of BERT's performance on the same benchmarks [9]. Distillation differs from other forms of model compression in that it involves training a new model, while generally other forms of compression modify a pre-existing model directly.

Pruning is a compression technique involving the removal of weights, nodes, or other elements from a model. Various heuristics have been developed to identify the elements which can be pruned with the least impact on overall performance. In this work, we focus on magnitude based pruning, which

removes the smallest weights with the assumption that they contribute the least to the output [10]. This has the advantage of being very computationally simple and easy to implement, but is by no means a state-of-the-art pruning algorithm. Pruning only weights and not larger structures does not provide easily-realized speedup [11]. Additionally, pruning without fine-tuning will eventually ruin a model’s performance completely due to accumulating errors. We expect that fine-tuning the pruned model would overwrite any effect that the pruning itself had on fairness, but this could certainly be an area for future research.

Quantization is one of the most popular compression techniques in large part due to its overall simplicity and large levels of achievable compression. It involves converting 32-bit floating point numbers used in neural networks to lower precisions formats like 8-bit integers. These lower precision representations offer drastic reductions in memory demand, and in the case of integer quantization utilize processor optimizations [12]. To quantize post training requires a mapping values from the original to the compressed value range, often determined using a scaling factor derived from calibration data. In general quantization results in slight decreases in model accuracy as the representations become smaller. However the level of accuracy lost differs greatly across different domains and models. For our work we focus mainly on integer based quantization for 16-bit, 8-bit, and 4-bit integer representations. With 8-bit integers being the most standard form of quantization.

IV. EXPERIMENTAL DESIGN (DATASETS/TASKS)

To begin we experimented with a fully connected neural network (FCNN) trained on tabular datasets COMPAS Recidivism and UCI Adults Income prediction [13], [14]. During the training process, an optional fairness constraint method [7] is implemented to equalize prediction rates across protected groups. We then use the previously discussed quantization, distillation, and pruning methods to compress the basic FCNN model and the fairness constrained model. For these models trained with tabular datasets, we used a fairness benchmarking script based on functions from AI Fairness 360 and Fairlearn to calculate demographic parity and disparate impact[15].

Computer vision models are frequently used in resource constrained edge devices where there is a need for fast inference with high levels of accuracy. To evaluate the effects of compression on computer vision models we use the FairFace dataset for binary gender classification. At the time of release, many of the popular computer vision datasets containing human faces were found to be racially biased. Datasets such as CelebA and UTKFace were primarily composed of white males and thus models trained using them tended to have reduced performances for other racial and gender groups [16]. The FairFace dataset aims to address this by providing a dataset with even distributions across 7 different race and 2 gender labels. In their work they found training a gender classifier on their dataset resulted in better accuracy for racial groups, however there was still some level of bias. For our experiments we replicate their work through both using their

base uniform distribution, and to model other datasets we create a rebalanced version, which oversamples faces with a race label of white by 60% and undersamples the other racial groups 40%. ResNet18 and ResNet50 models pretrained on ImageNet are then fine tuned for gender classification on the two datasets [17]. The fine tuned models are either quantized to 16-bit, 8-bit, and 4-bit precision or pruned by 10, 20, 30, 40, or 50%. Finally all models are evaluated on the same uniformly distributed test dataset.

Automatic speech recognition (ASR) systems are widely used in various applications, including virtual assistants, transcription services, and accessibility tools. To evaluate the fairness impacts of model compression on ASR systems, we experiment with the Whisper Tiny model. For our experiments, we apply quantization to reduce the bit precision of the model’s parameters to 16-bit, 8-bit, and 4-bit representations, in addition to maintaining the original 32-bit precision. We further perform pruning on both the uncompressed and quantized versions, gradually increasing pruning from 0% to 50% sparsity. These experiments allow us to investigate whether compression techniques exacerbate or mitigate biases in ASR performance across different speaker demographics. To assess fairness across language and gender, we use the Mozilla Common Voice 13 dataset, which contains transcribed speech samples labeled by speaker gender and language. We focus on male and female speaker splits across three different languages, yielding approximately 3,000 total samples for evaluation. We analyze the performance discrepancy between gender and language groups across different quantization levels and pruning percentages, measuring both overall transcription accuracy and performance disparities.

Language models are widely used in natural language processing (NLP) tasks such as text classification, machine translation, and conversational AI. Given their extensive applications, it is important to evaluate how compression techniques affect both performance and fairness in NLP systems. To investigate the effects of model compression on fairness in NLP, we experiment with quantization and model distillation across multiple widely used transformer-based masked language models (MLMs). Due to computational limitations, we do not manually distill models but instead rely on existing pre-distilled versions of popular baseline models. Specifically, we test BERT-Base and DistilBERT-Base, BERT-Large and MobileBERT, and RoBERTa and DistilRoBERTa, comparing their performance and fairness under compression. Each model undergoes quantization, reducing bit precision from 32-bit floating-point to 16-bit, 8-bit, and 4-bit integer representations. This allows us to analyze whether lower bit precision disproportionately affects specific demographic groups or biases model predictions. For evaluation, we use two well-established NLP bias benchmarking datasets: CrowS-Pairs, a dataset designed to measure biases in masked language models through paired sentences where one reflects a stereotype and the other a counter-stereotype. We also use StereoSet [18], a dataset that evaluates models on contextual biases by measuring preference for stereotypical, anti-stereotypical, and unrelated completions for given sentence prompts.

V. RESULTS

A. Fully Connected Neural Network

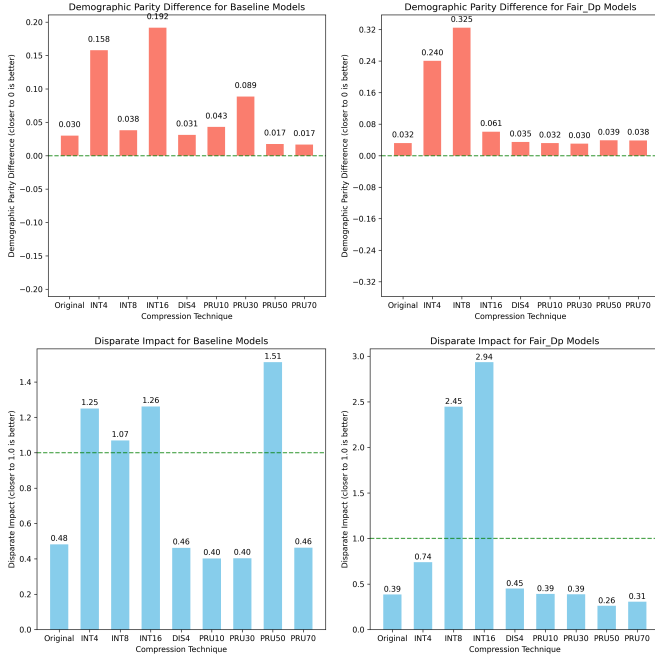


Fig. 1: Left and right: baseline and fairness constraint models. Top and bottom: Demographic Parity and Disparate Impact

For the case of FCNN models, we investigate the results of income prediction with the UCI Adult dataset with “sex” being the protected attribute. To compare the difference in positive prediction rates between privileged (male) and unprivileged groups (female), Demographic Parity Difference and Disparate Impact metrics are chosen for their suitability for this task and shown in Fig. 1, where the broken green lines indicate perfect fairness. Here we see a general trend of quantization gradually raising bias with higher bits, while bias after distillation and pruning remains at a similar level. In some pruning levels, the results show steep rises in bias or slightly increased fairness. However, in these cases, the pruned models have become effectively unusable as the accuracy has greatly declined compared to that of the baseline model. Last but not least, we see the fairness constraint mechanism consistently increases bias. This goes against its purpose and the parameters in the Binary Cross-Entropy loss function definitely require more tweaking.

B. ResNet

In the original FairFace paper the authors measure the fairness gaps through the maximum accuracy disparity which is defined as:

$$\varepsilon = \max \left(\log \left(\frac{\Pr(\hat{Y} = Y | A = j)}{\Pr(\hat{Y} = Y | A = k)} \right) \right) \quad (1)$$

where A is the protected attribute, $\hat{Y}$ the predicted gender, and Y is the true gender [16]. In our case the protected attributes would be either the race or gender of the image

classified. For our experiments we took the maximum of the maximum accuracy disparity across multiple runs to represent the worst case fairness disparity. The gender disparity and accuracy plots as we quantize the ResNet18/50 models fine tuned on the balanced and unbalanced datasets are shown in Fig. 2. Here we have omitted the quantization to 4-bit integers as the performance of the model drastically degrades and reaches a gender classification accuracy of roughly 50%. We can observe that the base gender disparity of our models is higher than those reported in the FairFace paper. This likely occurred because we are fine tuning a pretrained model on their dataset while in their work they train a ResNet34 model on the datasets from scratch. The race disparity is however in line with what they reported. Quantizing the ResNet models down to 16-bit integers there are very slight changes to both the maximum accuracy disparity and the overall accuracy of the model across the board. Quantizing to 8-bit integers, there are significant changes.

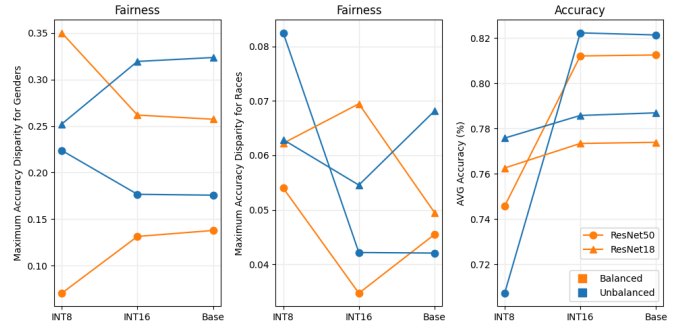


Fig. 2: Left and Middle: Maximum accuracy disparity over quantization for gender and race respectively. Right: Overall Accuracy over quantization

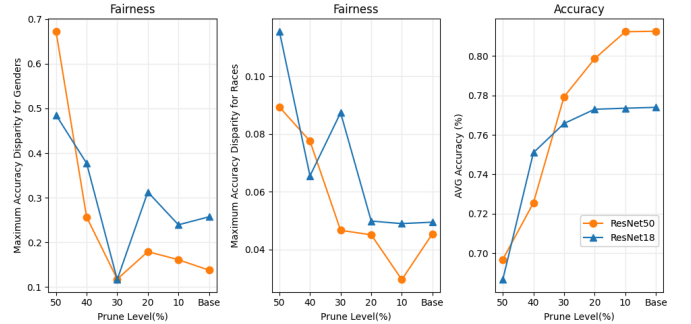


Fig. 3: Left and Middle: Maximum accuracy disparity over pruning level for gender and race respectively. Right: Overall Accuracy over pruning level

For all model versions there is a drastic change in maximum accuracy disparity with some seeing significant increases in fairness and others decreases. The resulting fairness does not exhibit any clear pattern, however this is also across a limited sample size. More samples may be needed to discern a true pattern.

In contrast repeating the above experiments on the balanced dataset using pruning in Fig. 3 we can see that there is a definite decrease in fairness as we increase the level of pruning from 10% to 50%. In the case of 30% pruning, the fairness for genders greatly improves before immediately getting worse at

40% pruning. Indicating pruning has a noticeable affect on the fairness of the model overall.

C. Whisper

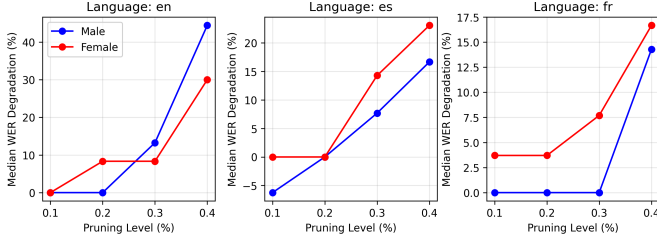


Fig. 4: Median WER degradation percentage for male and female speakers across increasing pruning levels (10% to 40%) in English, Spanish, and French. While English degrades more significantly at higher pruning levels, Spanish and French show inconsistent gender disparities, with women experiencing higher degradation in both languages.

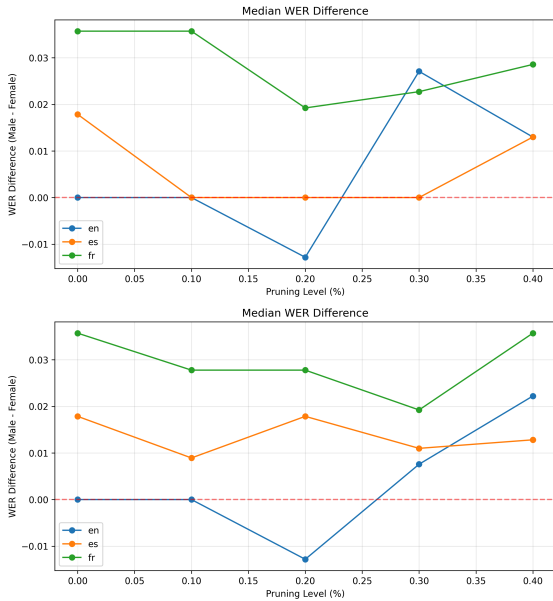


Fig. 5: Median WER for male and female speakers across pruning levels at 32-bit (top) and 8-bit (bottom) precision. A notable gender gap is seen in French when quantizing from 32 to 8-bit precision, highlighting potential fairness concerns introduced by quantization

To evaluate the impact of pruning on fairness in ASR, we analyze the median word error rate (WER) degradation percentage between male and female speakers across different pruning levels (10% to 40%) for English, Spanish, and French. The results are shown in Fig. 4. Across languages, we do not observe a consistent pattern in gender-based degradation. In Spanish and French, WER degradation is more pronounced for female voices, whereas in English, degradation is higher for male voices. This suggests that the impact of pruning on fairness varies by language, possibly due to differences in linguistic structure, pronunciation variations, or dataset composition. One interesting observation is that English degrades much more significantly at higher pruning levels, reaching a 30-40% WER degradation, whereas Spanish and French maintain relatively stable performance. Additionally, we observe an improvement in WER for male Spanish voices

at 10% pruning, indicating that mild pruning may act as a form of regularization.

In Fig. 5, we analyze the median WER of male and female speakers across pruning levels from 0% to 40%, comparing models at 32-bit and 8-bit precision. An interesting observation is that at 32-bit precision, WER remains relatively balanced between male and female speakers across Spanish, however, at 8-bit precision, we observe a distinct gap in WER for Spanish, where male observe higher error rates than female voices. This suggests that aggressive quantization disproportionately impacts gender fairness in certain languages, even when pruning levels remain constant. The emergence of this bias at lower bit precision highlights a potential trade-off between efficiency and fairness, particularly in multilingual ASR systems. These results emphasize the need for fairness aware compression strategies to mitigate unintended biases introduced through model compression. Future work could explore these behaviors across more languages and more fine grained pruning levels to smoothen out behaviors.

D. Masked Language Models

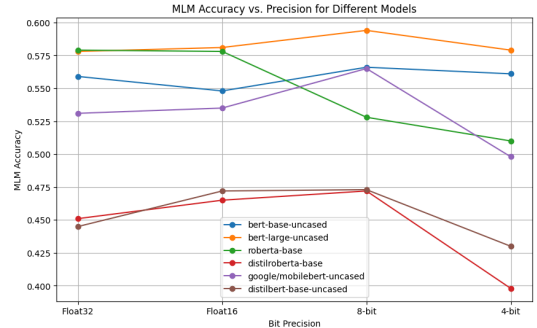


Fig. 6: Model Accuracy Across Distilled and Original Versions at Different Bit Precisions evaluated on GLUE

We first evaluate the general performance accuracy on the standard GLUE dataset [19] of both distilled and original masked language models (MLMs) at varying bit precisions. The models analyzed performance accuracy of BERT-Base, BERT-Large, and RoBERTa compared to their respective distilled versions—DistilBERT-Base (distilled from BERT-Base), MobileBERT (distilled from BERT-Large), and DistilRoBERTa (distilled from RoBERTa). As shown in Fig. 6, across all tested models, the distilled versions consistently have lower accuracy compared to their original counterparts at full 32-bit precision. However, the distilled models improve accuracy at 8 and 16 bit precisions, whereas the original models either maintain or degrade in accuracy.

To analyze the fairness effects of quantization, we measure bias using the StereoSet Bias Score (SSS) across models and bit precisions. The StereoSet bias score [20] is computed using the following equation:

$$\text{sss}(s) := \frac{1}{l_s} \sum_{t \in s} \log P(t|U; \theta) \quad (2)$$

Where $\text{sss}(s)$ represents the StereoSet score for a given sentence s , and $P(t|U; \theta)$ denotes the probability assigned by the model to a given token t given its context U . From Fig. 7,

we observe that reducing bit precision generally leads to a slight reduction in bias, with 4-bit precision models showing the lowest bias across all tested configurations.

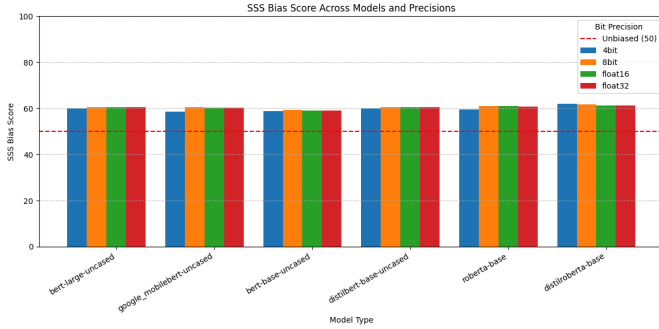


Fig. 7: Impact of Bit Precision on Fairness Measured by StereoSet Bias Score (SSS) across different bit precisions for all models

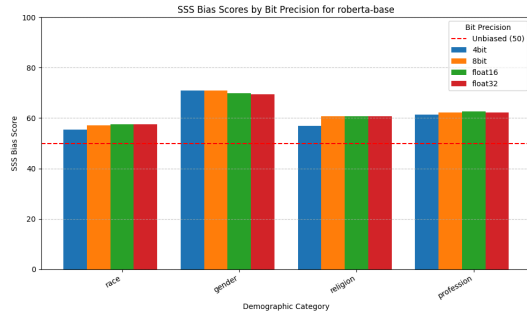


Fig. 8: Per-Class Fairness Impact of Quantization Across Bit Precisions. Analysis of StereoSet Bias Scores (SSS) across different protected classes for RoBERTa-base at varying bit precision

Building on these results, Fig. 8 investigates how quantization affects fairness within different protected classes using RoBERTa as an example.

Most classes exhibit improved fairness when quantizing to lower bit precision. However, some classes see slight increases in bias, and these effects are not consistent across different models with different models having different classes sometimes increase bias. While 4-bit precision generally reduces bias, it remains unclear whether this effect is due to a true unlearning of bias or simply a reduction in overall model capability. These findings suggest that while quantization can help mitigate bias, its impact varies depending on the specific model, bit precision, and demographic group. Further investigation is needed to determine whether this fairness improvement is an inherent potential benefit of quantization or merely an artifact of performance degradation.

VI. CONCLUSION

Across different models and applications, a small amount of pruning and quantization showed some impact on fairness metrics, with weight based pruning having a more limited benefit before it started to have varying effects on accuracy and fairness. Alternative forms of pruning and the addition of post-training could likely extend its usefulness. Quantization’s impact on fairness at higher compression ratios showed more

promise, but still had a negative impact on fairness in some cases and we this improvement always came with a decrease in overall accuracy. Distilling seemingly had the least impact on fairness, however there are still many avenue to be explored.

Because we observed significant effects on fairness in some cases when more than a light amount of compression was applied, we encourage developers using these techniques to keep an eye on fairness metrics and not just on overall accuracy. For those hoping that compression can be a source of increased fairness, case-by-case testing may show some improvements with pruning and quantization, but this is far from guaranteed.

VII. LIMITATIONS AND FUTURE WORK

Inherently working with model compression involves using large models that demand substantial levels of computational power which we had limited access to. As a result we were also unable to use pure knowledge distillation and instead relied on pre-distilled versions of models that were available. The computational demand also meant we were limited in the size of models used for instance using ResNet50 instead of ResNet100. In addition, we initially planned to evaluate our computer vision task on automatic facial expression recognition which is known to result in racial biases [21]. However we were unable to acquire access to the relevant FER datasets and their corresponding protected labels after reaching out to the authors.

Potential future avenues of work could include exploring compressing FER models and publicly available large language models such as DeepSeek. LLMs are some of the largest models and they therefore stand to benefit the most from model compression. However, until recently, cutting edge LLMs tended to be closed source and therefore not available for compression research, so the effects on fairness of these LLMs is not fully explored. Additionally, using pure knowledge distillation could also provide for a direct level of comparison between different “teacher” models and their compressed counterparts. Another avenue to pursue could be how fairness-aware models’ fairness is affected by compression. In our work we were only able to touch on simpler examples for COMPAS and ADULT, and not more complex tasks such as ASR, NLP, or CV. Finally, at this moment in time fairness in automatic speech recognition is a relatively underexplored topic relative to fairness in compression in computer vision and natural language processing. A deeper dive into model compression’s effects on ASR model’s fairness could prove to be both interesting and novel for future researchers.

REFERENCES

- [1] S. Hooker, N. Moorosi, G. Clark, S. Bengio, and E. Denton, “Characterising Bias in Compressed Models.” [Online]. Available: <https://arxiv.org/abs/2010.03058>
- [2] S. Stoychev and H. Gunes, “The Effect of Model Compression on Fairness in Facial Expression Recognition,” in *Pattern Recognition, Computer Vision, and Image Processing. ICPR 2022 International Workshops and Challenges*, J.-J. Rousseau and B. Kapralos, Eds., Cham: Springer Nature Switzerland, 2023, pp. 121–138.
- [3] X. Lin, S. Kim, and J. Joo, “FairGRAPE: Fairness-Aware GRAdient Pruning mEthod for Face Attribute Classification,” in *Computer Vision*

- *ECCV 2022*, S. Avidan, G. Brostow, M. Cissé, G. M. Farinella, and T. Hassner, Eds., Cham: Springer Nature Switzerland, 2022, pp. 414–432.
- [4] G. Xu and Q. Hu, “Can Model Compression Improve NLP Fairness.” [Online]. Available: <https://arxiv.org/abs/2201.08542>
 - [5] K. Ramesh, A. Chavan, S. Pandit, and S. Sitaram, “A Comparative Study on the Impact of Model Compression Techniques on Fairness in Language Models,” in *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, A. Rogers, J. Boyd-Graber, and N. Okazaki, Eds., Toronto, Canada: Association for Computational Linguistics, Jul. 2023, pp. 15762–15782. doi: 10.18653/v1/2023.acl-long.878.
 - [6] T. P. Ferraz, “Efficient Compression of Multitask Multilingual Speech Models.” [Online]. Available: <https://arxiv.org/abs/2405.00966>
 - [7] M. B. Zafar, I. Valera, M. G. Rodriguez, and K. P. Gummadi, “Fairness constraints: Mechanisms for fair classification,” in *Artificial intelligence and statistics*, 2017, pp. 962–970.
 - [8] G. Hinton, O. Vinyals, and J. Dean, “Distilling the knowledge in a neural network,” *arXiv preprint arXiv:1503.02531*, 2015.
 - [9] V. Sanh, L. Debut, J. Chaumond, and T. Wolf, “DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter,” *arXiv preprint arXiv:1910.01108*, 2019.
 - [10] S. Vadera and S. Ameen, “Methods for Pruning Deep Neural Networks,” *IEEE Access*, vol. 10, no. , pp. 63280–63300, 2022, doi: 10.1109/ACCESS.2022.3182659.
 - [11] H. Cheng, M. Zhang, and J. Q. Shi, “A Survey on Deep Neural Network Pruning: Taxonomy, Comparison, Analysis, and Recommendations,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 46, no. 12, pp. 10558–10578, 2024, doi: 10.1109/TPAMI.2024.3447085.
 - [12] H. Wu, P. Judd, X. Zhang, M. Isaev, and P. Micikevicius, “Integer quantization for deep learning inference: Principles and empirical evaluation,” *arXiv preprint arXiv:2004.09602*, 2020.
 - [13] J. Angwin, J. Larson, L. Kirchner, and S. Mattu, “Machine bias.” [Online]. Available: <https://www.propublica.org/article/machine-bias-risk-assessments-in-criminal-sentencing>
 - [14] B. Becker and R. Kohavi, “Adult.” 1996.
 - [15] R. K. E. Bellamy *et al.*, “AI Fairness 360: An Extensible Toolkit for Detecting, Understanding, and Mitigating Unwanted Algorithmic Bias.” [Online]. Available: <https://arxiv.org/abs/1810.01943>
 - [16] K. Kärkkäinen and J. Joo, “Fairface: Face attribute dataset for balanced race, gender, and age,” *arXiv preprint arXiv:1908.04913*, 2019.
 - [17] K. He, X. Zhang, S. Ren, and J. Sun, “Deep Residual Learning for Image Recognition,” in *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770–778. doi: 10.1109/CVPR.2016.90.
 - [18] M. Nadeem, A. Bethke, and S. Reddy, “StereoSet: Measuring stereotypical bias in pretrained language models,” *arXiv preprint arXiv:2004.09456*, 2020.
 - [19] A. Wang, A. Singh, J. Michael, F. Hill, O. Levy, and S. R. Bowman, “GLUE: A multi-task benchmark and analysis platform for natural language understanding,” *arXiv preprint arXiv:1804.07461*, 2018.
 - [20] R. Zalkikar and K. Chandra, “Measuring Social Biases in Masked Language Models by Proxy of Prediction Quality,” *arXiv preprint arXiv:2402.13954*, 2024.
 - [21] M. M. Hosseini, A. P. Fard, and M. H. Mahoor, “Faces of Fairness: Examining Bias in Facial Expression Recognition Datasets and Models,” *arXiv preprint arXiv:2502.11049*, 2025.